



NAME:.....ADM:.....

MATHEMATICS
FORM 2
TIME: 1 HOUR 15 MINS

1. Use tables to evaluate.

(3mks)

$$\sqrt{\left(\frac{3.45 \times 16.7}{31.5} \right)}$$

2. Solve for x in each of the following equations.

(3mks)

(a) $3^{(2x-5)} = 27$

(b) $3^{4x} \div 3^{-7} = 3^{15}$

(3mks)

3. Use reciprocals tables to evaluate

(3mks)

4. A metallic cuboid measuring 16cm by 8cm by 4 cm was melted . The material was then used to make a cube. What was the length of the cube?

(3mks)



5. Simplify

$$\sqrt{\frac{27x^3y^9}{x^6y^3}}$$

(3mks)

6. Find the equation of the line through the points A (2, 5) and B(3, 11)

(3mks)

7. Determine the equation of the line perpendicular to the line whose equation is $y = -5x + 3$ and passes through the point (3, 2). (3mks)

8. A(-5, -2), B(-2, -5) and C(-12, -2) are vertices of a triangle. Find the image of the triangle when it is reflected in :

(a) The line $y = -x$ (4mks)

(b) The line $y = x$ (4mks)



9. Find the area in hectares of a coffee field whose measurements are entered in a field book as shown below. Take $xy=200\text{m}$ as the baseline. (8mks)

	Y	
	180	40 to Q
To R 80	140	
To S 160	100	
	40 100 to P	
	X	

10. Use the reciprocal tables and square root to evaluate. (4mks)

$$\frac{0.1 + 0.498}{0.0351} \sqrt{\quad}$$

11. Two men each working for 8 hours a day can cultivate an acre of land in 4 days. How long would 6 men each working in 4 hours a day take to cultivate 4 acres? (3mks)

12. The sum of interior angles of a regular polygon is 1080° . Find the size of each exterior angle. (3mks)